ClonoDist enables deterministic, reference-qualified T-cell receptor distance and neighborhood analysis

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Abstract

Reproducible T-cell receptor analysis requires a defined distance method, explicit reference context, and inspectable downstream decisions. ClonoDist connects these requirements through complementary Core and Extended layers without introducing a new distance formula. Core preserves qualified human and mouse alpha/beta computation, whereas Extended introduces qualified gamma/delta contexts, explicit execution profiles, and a decoupled workflow for meta-clonotype discovery, operational qualification, and application. Computational tests establish exact integer parity with the fixed Python reference within the declared domains; correlation displays illustrate this equality rather than serve as its formal test. On an Apple M4 Pro platform, complete-process medians for two synthetic dense workloads were 0.076947 and 0.098113 seconds for ClonoDist, versus 2.530671 and 2.561100 seconds for Python. A study-separated human TRB evaluation yields descriptive operational matches with explicit controls and rejection reasons. These results support traceable computation and inspectable workflows, not antigen prediction, population error calibration, or clinical validity.

Introduction

Quantifying similarity between T-cell receptors depends on multiple interdependent conventions: sequence annotations, germline reference databases, regional boundaries, trimming parameters, gap penalties, and distance aggregation across complementarity-determining regions. Because these conventions dictate numerical outcomes, reproducing a receptor-distance method requires substantially more than implementing its pairwise alignment arithmetic. Downstream neighborhood analyses introduce further operational choices—selecting cluster centers, defining search radii, evaluating background repertoires, and applying acceptance thresholds—that must likewise remain fully inspectable.

Rather than serving merely as a compiled language port, ClonoDist provides an auditable execution and analysis framework built upon the established tcrdist3/pwseqdist scoring method [1, 2, 6, 7]. ClonoDist introduces no new receptor-distance formula and does not claim that sequence proximity implies shared antigen recognition. Instead, it addresses the broader challenge of analytical reproducibility: enforcing reference-aware sequence admission, binding explicit execution profiles, producing deterministic scientific outputs within qualified execution settings, and recording unambiguous reasons for operational clustering decisions.

ClonoDist organizes these capabilities into two complementary layers: Core preserves the qualified human and mouse alpha/beta distance engine, while Extended uses this shared kernel to introduce qualified gamma/delta receptor contexts, configurable execution profiles, and a decoupled meta-clonotype pipeline. These qualification layers remain distinct: exact scoring, configuration mechanics, and observational workflow behavior do not establish equivalent biological validity. Here, we present three connected results: first, how reference qualification formalizes scoring conventions and expands receptor scope; second, exact computational parity against the Python reference alongside local runtime performance; and third, an inspectable operational neighborhood analysis that tracks selection criteria and rejection reasons in study-separated human repertoires.

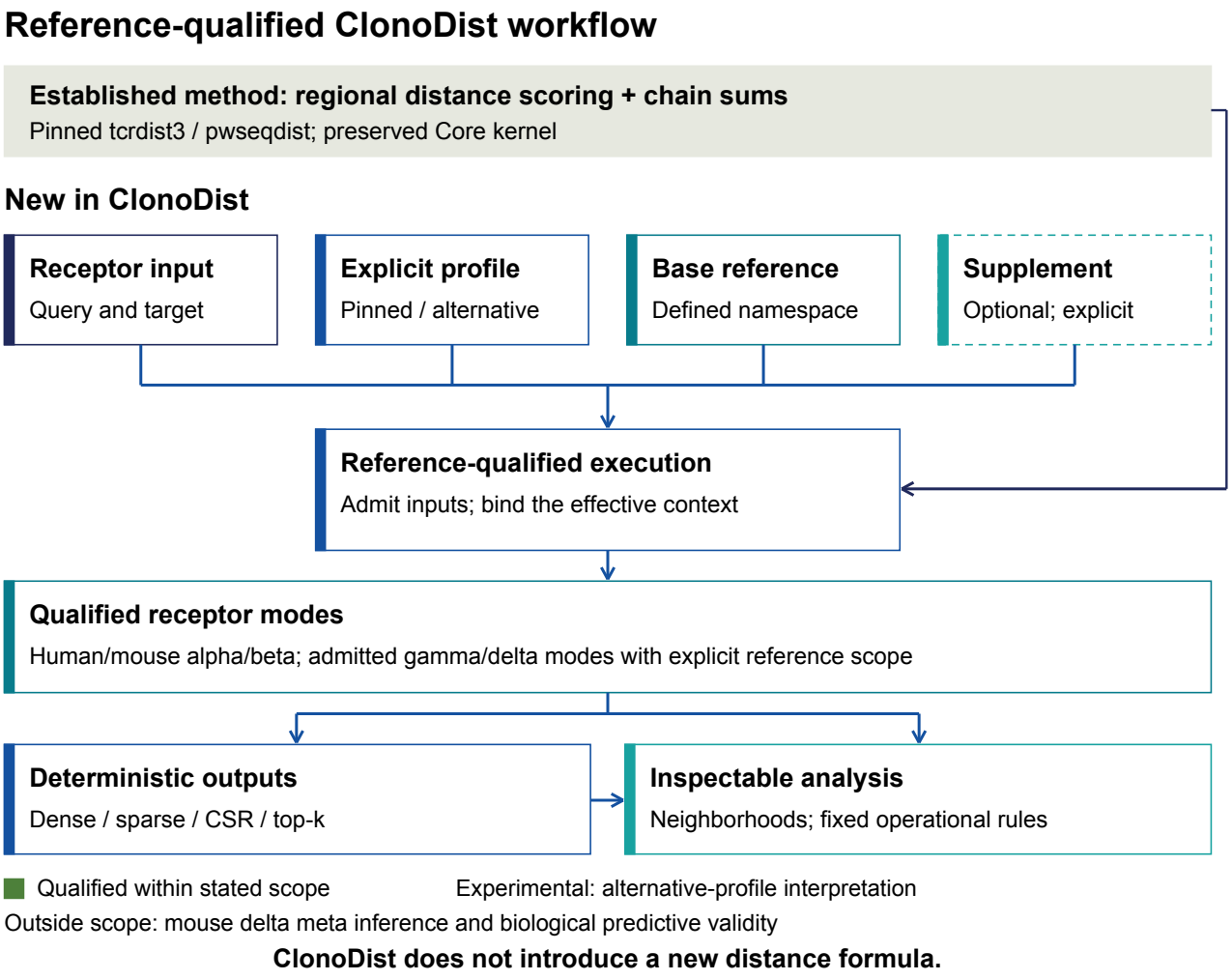
Results

Reference-qualified execution connects established scoring to inspectable outputs

Beyond translating scoring algorithms into a compiled language, ClonoDist formalizes the computational dependency chain by binding input data, parameter profiles, and germline reference databases into an explicit execution context (Figure 1). Within qualified modes, ClonoDist preserves established tcrdist3/pwseqdist regional scoring rules and chain sums; Core behavior remains unchanged, and no new distance formula is introduced.

Within this framework, Core evaluates human and mouse TRA, TRB, and paired TRA+TRB chains. Extended expands receptor scope to include human TRG, dedicated human TRD, paired TRG+TRD, mouse TRG, and explicit supplemental mouse TRD and paired distances. To ensure unambiguous germline mapping, the mouse delta supplement admits exactly 51 V and three J alleles in a defined namespace, avoiding universal strain extrapolation and preventing silent admission of shared human alpha/delta alleles (Supplementary Figure S1). Biological interpretation of alternative weighting profiles remains experimental; tested execution settings and output semantics appear in Methods and Supplementary Figure S2.

Figure 1



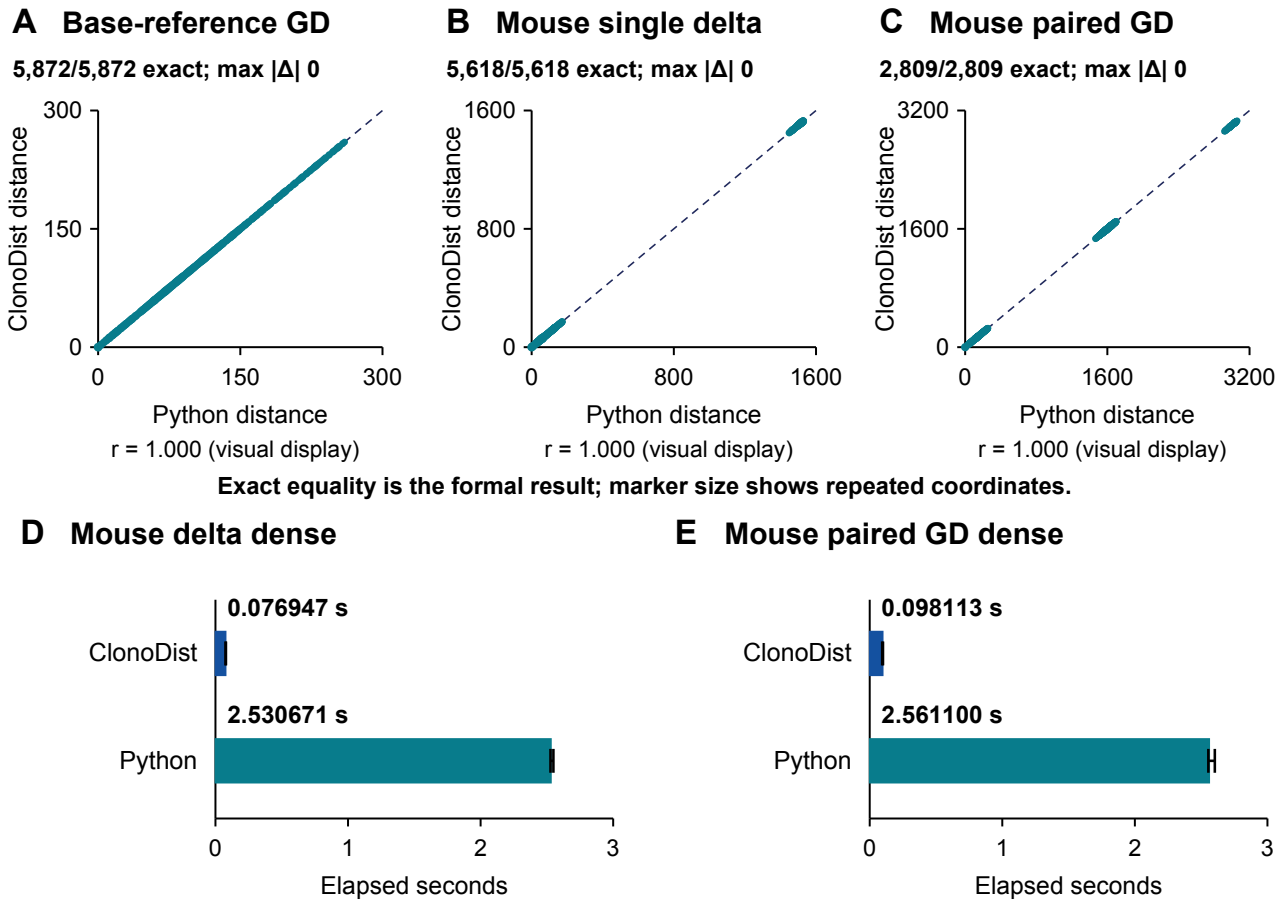
Connected ClonoDist workflow. Inputs, an explicit profile, the base reference and any optional supplement define reference-qualified execution. Qualified receptor modes feed deterministic outputs and inspectable neighborhood analysis. The established tcrdist3/pwseqdist scoring method and preserved Core kernel remain the foundation; the connected execution and analysis framework is the ClonoDist contribution. Core and Extended are complementary. The supplement is explicit, and qualification is limited to the admitted reference namespace. Alternative-profile biological interpretation is experimental; mouse delta meta inference is outside scope. ClonoDist does not introduce a new distance formula.

Exact parity and local runtime provide separate evidence

Across qualified receptor domains, ClonoDist reproduces fixed Python reference totals for all 5,872 original gamma/delta and 8,427 mouse pairs with zero maximum absolute difference (Figure 2). Visual correlation displays illustrate this agreement; exact equality is the formal parity result. As detailed in Methods, displayed native values reconstruct previously verified outputs rather than represent a new scoring run or independent observations.

Runtime was evaluated alongside parity as separate operational evidence. For synthetic delta/paired dense workloads, ClonoDist medians were 0.076947/0.098113 seconds versus Python's 2.530671/2.561100 seconds. Horizontal bars summarize three complete new-process observations with observed min–max whiskers, not confidence intervals. These local Apple M4 Pro measurements do not establish real-mouse scaling or universal speed; full conditions and limitations appear in Methods.

Figure 2



Scientific parity and local runtime. A–C show visual correlation of 5,872, 5,618 and 2,809 directed pair entries; marker size represents repeated coordinates without jitter. Every entry matches exactly and maximum absolute difference is zero. Exact equality is the formal result. Native values reconstruct previously verified output files; no new native scoring was performed. D–E show horizontal median bars and observed minimum-to-maximum whiskers from three complete new-process observations per summary; whiskers are not confidence intervals. Each workload is synthetic and dense (1,000 × 1,000) on the recorded Apple M4 Pro platform. Setup, serialization and wrapper differences are included. Full conditions and limitations appear in Methods; these observations do not establish biological validity, real-mouse scaling or performance on other platforms.

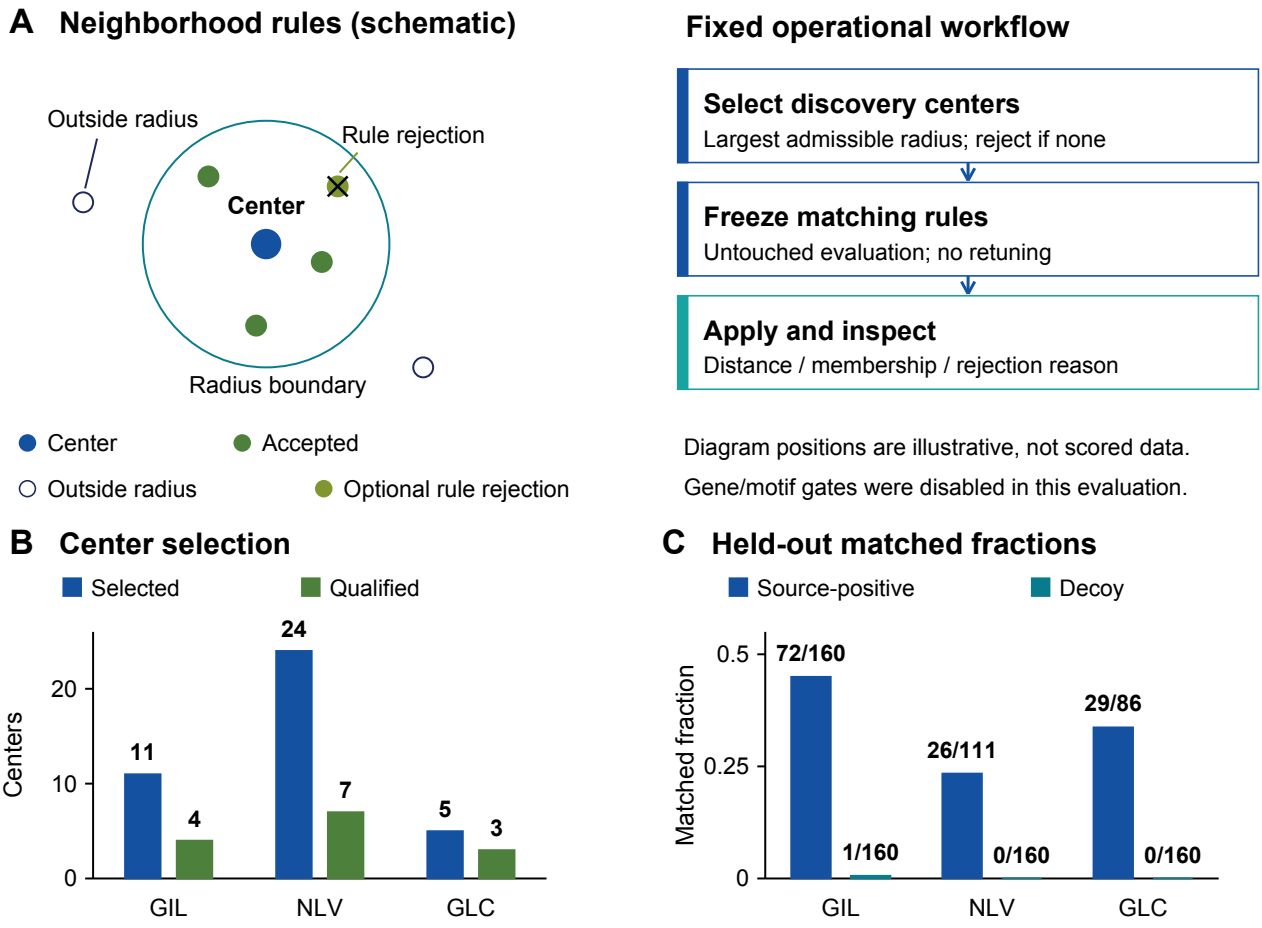
Operational matching retains selection rules and decision reasons

Beyond pairwise distances, Extended formalizes neighborhood analysis by fixing discovery and matching rules before untouched evaluation and retaining decision reasons (Figure 3). The neighborhood illustration is schematic; optional gene/motif gates were disabled in this evaluation, and distance zero is not direct identity.

Pinned-profile selected and qualified centers numbered 11/4, 24/7 and 5/3 for GILGFVFTL, NLVPMVATV and GLCTLVAML. Source-positive versus decoy matches were 72/160 versus 1/160, 26/111 versus 0/160 and 29/86 versus 0/160. These are descriptive matches, not sensitivity/specificity or inferred HLA restriction.

Controls accepted 16/16 synthetic positives and 0/16 distant negatives. The experimental CDR3-focused NLV reversed-label control retained four centers, limiting its interpretation (Supplementary Figure S3). Operational matches and background bounds do not establish biological predictive validity.

Figure 3



GIL: GILGFVFTL NLV: NLVPMVATV GLC: GLCTLVAML

Computational controls and profile sensitivity: Supplementary Figure S3

Operational matching does not establish biological predictive validity.

Neighborhood semantics and operational matches. A is a schematic: a center has accepted neighbors, outside-radius receptors and an illustrative within-radius optional-rule rejection. Positions are not receptor measurements. Gene/motif gates were disabled in this evaluation. The fixed workflow selects centers, evaluates without retuning and retains decision reasons. B shows selected and operationally qualified centers; C shows descriptive source-positive/decoy matched fractions for study-separated human TRB. These are not sensitivity or specificity estimates. Imperfect decoys and donor ambiguity preclude population calibration. Operational matching does not establish antigen prediction, inferred HLA restriction, causal biology or clinical validity. Controls and experimental-profile limitations are retained in S3 and Methods.

Discussion

ClonoDist contributes more than a language translation by formalizing reference admission, profile identity, output semantics and operational decision rules. Rather than modifying the distance definition, these additions establish an auditable execution framework that connects exact arithmetic parity to reproducible analysis.

Reference choice is an integral component of scientific context. The mouse delta analysis shows why generic database defaults cannot replace explicit curation: subtle allele differences alter computed distances. Explicit shared-allele admission, namespaces and preserved exclusions ensure inspectability without extrapolating to unexamined alleles or strains.

Operational matching introduces downstream choices—radii, backgrounds and filtering thresholds—that must remain inspectable without treating qualification as biological certification. The retention of four centers under the experimental CDR3-focused negative control illustrates why configurable mechanics do not establish biological specificity; no untouched holdout remains to evaluate that alternative.

The recorded timings demonstrate practical local throughput for the declared synthetic workloads without asserting universal language speedups or large-scale scaling guarantees. By separating verified parity and inspectable operational mechanics from uncalibrated biological inference, ClonoDist provides a transparent framework for qualified T-cell receptor distance and neighborhood analysis.

Methods

Implementation and execution context

All analyses use fixed ClonoDist Core and Extended implementations and tcrdist3 0.3 and pwseqdist 0.6 source versions [1, 2, 8]. The Python reference environment included Parasail, whose alignment library is described by Daily [9]. Core behavior is preserved in Extended. The parity displays reconstruct existing numerical results; no additional scoring or benchmark observations were generated for this manuscript.

Strict admission requires exact V/J alleles, independently supplied canonical CDR3 sequences of length 1–127, stable IDs and positive checked supplied counts. Counts retain their supplied units rather than inferred cell or molecule units. The effective profile is defined independently of its file formatting; input, profile, base reference and optional supplement jointly define the execution context.

The pinned profile uses CDR1, CDR2 and analytical pmhc/CDR2.5 outer weights of one and CDR3 weight three, with inherited substitution, gap, trimming and chain-sum rules. Custom and CDR3-focused profiles have qualified configuration mechanics but experimental biological interpretation. Radius thresholds are inclusive; normalization preserves supplied-count totals. Missing sparse coordinates mean absent edges; explicit zero edges remain distinct. Top-k ties follow distance then target ID; CSR indices are sorted.

References and parity reconstruction

The base database is the archived combo reference dated 2024-03-05. Mouse delta requires an explicit supplement derived from IMGT nucleotide snapshots retrieved on 2026-10-29 [3, 10]. Its 51 admitted V alleles comprise 11 dedicated and 40 explicitly shared alleles. Of 53 primary TRDV records, TRAV21/DV12*03 is excluded as ORF and TRAV6-7/DV9*08 for missing CDR1. Three TRDJ alleles are separately catalogued. F/(F)/[F] are retained source qualifiers, not independent functional or genomic validation. Feature-array intervals 28–39, 57–66 and 82–88 are not generic IMGT numbering; pmhc/CDR2.5 is an analytical feature.

Independent Python reference matrices preceded native comparison, with compiled pwseqdist regional crosschecks. Original gamma/delta tests cover 5,872 pair totals and 28,672 regional values; the three 53-record mouse cases cover 8,427 totals and 44,944 regional values. Core tests additionally include 192 receptor-pair cases, 256 chain scores, 1,024 components, 64 paired sums and 6,000 seeded regional probes. These overlapping categories are not pooled as independent samples.

For Figure 2, stored Python reference matrices were serialized in the original native dense-output format. All seven reconstructed gamma/delta and mouse files were verified as byte-for-byte identical to the previously recorded native outputs using their retained file identities. Native values were then parsed from those reconstructed files. This reconstructs the previously verified outputs and does not constitute a new native scoring run or independent measurements. Exact integer equality and maximum absolute difference zero are formal parity results. Directed entries include diagonals and duplicates. Correlation summarizes the visual alignment only.

Current-primary versus historical mouse-reference comparison found 200 differing directed totals among 49 records, entirely explained by TRDV5 CDR2 source differences. These are reference-context differences, not unexplained native errors (Supplementary Figure S1).

Qualification also covers normalized identities, thresholds, sparse zeros, top-k, summaries and refusals. Mouse pinned modes test workers 1/2/4/8 with tiles 1/7/64 throughout the Cartesian grid. Original Extended pinned dense checks use only (1,1), (2,7), (4,64) and (8,7); Core radius-edge settings also differ (Supplementary Figure S2). Untested settings are not treated as passing. Determinism applies to scientific output files, excluding per-run timing and build metadata. Model integrity checks establish consistency rather than authenticated provenance.

Complete-process timing

Outer elapsed-time observations cover whole timed subprocesses, not rounded resource counters or internal compute time. Figure 2 uses each series' three observations to calculate median, minimum and maximum; whiskers are observed ranges, not confidence intervals. Scientific dense outputs match within each workload. Fixed executables, inputs, profiles and references were used for each comparison.

Both workloads repeat qualified variable-length fixture rows to 1,000 unique IDs and produce dense 1,000-by-1,000 output. Recorded hardware is Apple M4 Pro/Mac16,11 with 24 GiB RAM, macOS 27.0.1 ARM64, CPython 3.11.16 and Rust 1.98.0. Native defaults are one worker, 32-row tiles and a 1 GiB planning budget; Python uses one CPU. Python includes imports and Numba initialization; native adds provenance work. Filesystem caches were not flushed, and native/Python campaigns were not interleaved. All three observations in each recorded series are included. These wrapper-specific local results do not establish real mouse repertoire scaling, universal language speed or another platform's performance.

Operational discovery and evaluation

VDJdb 2025-12-29 human TRB source observations supply three prespecified epitope labels and HLA-A*02:01 annotation [4, 11]. Whole sorted studies alternate between discovery and evaluation; repeated V/CDR3 signatures are purged globally. Deterministic role caps are 160 records, with actual role sizes retained: NLV discovery decoys number 74. Study-qualified donor identifiers cannot exclude cross-study person recurrence.

Discovery centers come from foreground only. Fixed rules use radii 0/12/24/36/48/60, empirical background cap 0.01, minimum support three, Jaccard cap 0.8 and 64 donor-block resamples with seed 2901026. The largest admissible radius is selected; no-valid-radius candidates are explicitly rejected, including contaminated zero-radius cases. Smoothed-enrichment and stable-ID ranking precede redundancy filtering. Models and rules are fixed before untouched evaluation; no evaluation retuning occurs.

Operational qualification additionally requires positive foreground-over-background prevalence, the empirical cap, at least two supporting held-out donors and exact-radius retention at least 0.6. Gene/motif gates were disabled in this evaluation but exercised in controls. Candidate-wise Fisher and zero-event binomial summaries are exploratory, not population-calibrated gates. Subsequent verification reproduced 18 retained models and the original application counts. Original application files were no longer retained, so their byte-for-byte identity was not assessed.

Background bounds for 28 retained centers use 1,034 contigs from one unlabeled 10x donor [5]. The 2,945 reference-possibility rows are metadata, not the fraction denominator. Bounds represent annotation possibilities, not confidence intervals or assigned genotypes; gene-constrained interpretation is unsupported. The donor has no established HLA or antigen-negative truth.

Limitations

Qualification is confined to the declared inputs, profiles, references, workflows and local environment. Computational parity and operational matching do not establish antigen prediction, inferred HLA restriction, causal biology, biomarker or clinical validity, or calibrated population false-positive control. General gamma/delta antigen inference and mouse delta meta inference remain outside scope. Synthetic cases are computational probes; public TRB evaluation is observational and uses imperfect decoys. Donor ambiguity and the single unlabeled background limit interpretation.

Custom, learned and CDR3-focused biological interpretations remain experimental. Local Apple Silicon qualification does not certify other platforms, hard process-RSS or OOM guarantees, or power-loss recovery. Managed allocation plans are not enforced RSS ceilings. Dense work/output may remain quadratic, and pruning depends on data and radius. Rights to use reference data, redistribute reference-derived material and distribute software are distinct; computational qualification does not establish redistribution permission.

Data and code availability

The ClonoDist source code and the supporting analysis materials are maintained in a fixed local repository. Figure-data tables and a reproducibility note accompany this manuscript. No public repository URL or archive accession has been assigned. Source code and supporting materials may be requested from the corresponding author, subject to applicable third-party terms.

Reproduction requires the stated software environment and the original reference and input material. The manuscript package is not a self-contained distribution of all external data. IMGT primary data and metadata carry a recorded CC BY 4.0 notice distinct from the terms for IMGT tools; that notice does not resolve the status of every historical reference-derived artifact. VDJdb-derived tables and models require separate consideration of the recorded CC BY-ND 4.0 terms. Reference use, redistribution and software licensing are separate matters; no unrestricted redistribution rights are asserted here.

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Competing interests

The author declares no competing interests.

References

[1] Mayer-Blackwell K, Schattgen S, Cohen-Lavi L, Crawford JC, Souquette A, Gaevert JA, Hertz T, Thomas PG, Bradley P, Fiore-Gartland A. TCR meta-clonotypes for biomarker discovery with tcrdist3 enabled identification of public, HLA-restricted clusters of SARS-CoV-2 TCRs. eLife. 2021;10:e68605. doi:10.7554/eLife.68605. https://elifesciences.org/articles/68605

[2] Fiore-Gartland A. pwseqdist: efficiently computes distances between protein sequences. Software, version 0.6. https://github.com/agartland/pwseqdist (accessed 2026-11-04).

[3] Lefranc M-P, Giudicelli V, Duroux P, Jabado-Michaloud J, Folch G, Aouinti S, Carillon E, Duvergey H, Houles A, Paysan-Lafosse T, Hadi-Saljoqi S, Sasorith S, Lefranc G, Kossida S. IMGT®, the international ImMunoGeneTics information system® 25 years on. Nucleic Acids Research. 2015;43(Database issue):D413–D422. doi:10.1093/nar/gku1056. https://pubmed.ncbi.nlm.nih.gov/25378316/

[4] Shugay M, Bagaev DV, Zvyagin IV, et al. VDJdb: a curated database of T-cell receptor sequences with known antigen specificity. Nucleic Acids Research. 2018;46(D1):D419–D427. doi:10.1093/nar/gkx760. https://pubmed.ncbi.nlm.nih.gov/28977646/

[5] 10x Genomics. Human T cells from a Healthy Donor, 1k cells - multi (v2). Single Cell Immune Profiling Dataset by Cell Ranger v5.0.0. 2020 November 19. https://www.10xgenomics.com/datasets/human-t-cells-from-a-healthy-donor-1-k-cells-multi-v-2-2-standard-5-0-0 (accessed 2026-11-04).

[6] Dash P, Fiore-Gartland AJ, Hertz T, et al. Quantifiable predictive features define epitope-specific T cell receptor repertoires. Nature. 2017;547(7661):89–93. doi:10.1038/nature22383. https://pubmed.ncbi.nlm.nih.gov/28636592/

[7] Mayer-Blackwell K, Fiore-Gartland A, Thomas PG. Flexible Distance-Based TCR Analysis in Python with tcrdist3. Methods in Molecular Biology. 2022;2574:309–366. doi:10.1007/978-1-0716-2712-9_16. https://pmc.ncbi.nlm.nih.gov/articles/PMC9719034/

[8] Mayer-Blackwell K and tcrdist3 contributors. tcrdist3: flexible CDR based distance metrics. Software, version 0.3. https://github.com/kmayerb/tcrdist3 (accessed 2026-11-04).

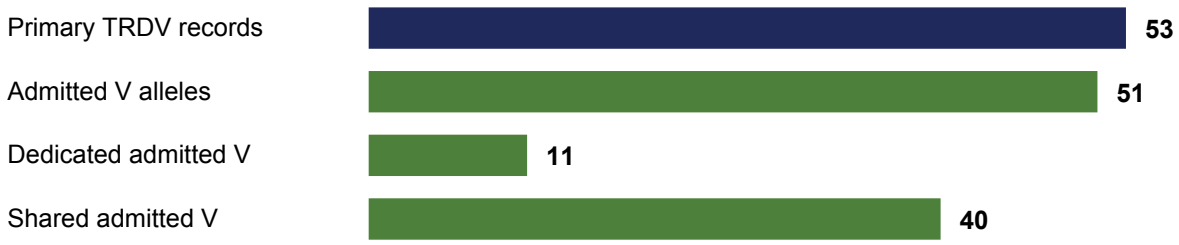
[9] Daily J. Parasail: SIMD C library for global, semi-global, and local pairwise sequence alignments. BMC Bioinformatics. 2016;17:81. doi:10.1186/s12859-016-0930-z. https://pubmed.ncbi.nlm.nih.gov/26864881/

[10] IMGT®, the international ImMunoGeneTics information system®. Mouse TRDV/TRDJ reference nucleotide sequences and allele annotations. Archived IMGT source snapshot: 2026-10-29. Reference directory: https://www.imgt.org/download/V-QUEST/IMGT_V-QUEST_reference_directory/Mus_musculus/TR/ . TRDV allele annotations: https://www.imgt.org/IMGTrepertoire/Proteins/taballeles/mouse/TRD/TRDV/Mu_TRDVall.html . Terms of use: https://www.imgt.org/about/termsofuse.php (accessed 2026-11-04).

[11] VDJdb Developers and Maintainers. VDJdb database, release 2025-12-29 (Happy New Year! release). Published 2025-12-30. https://github.com/antigenomics/vdjdb-db/releases/tag/2025-12-29 (accessed 2026-11-04).

Supplementary Figure S1

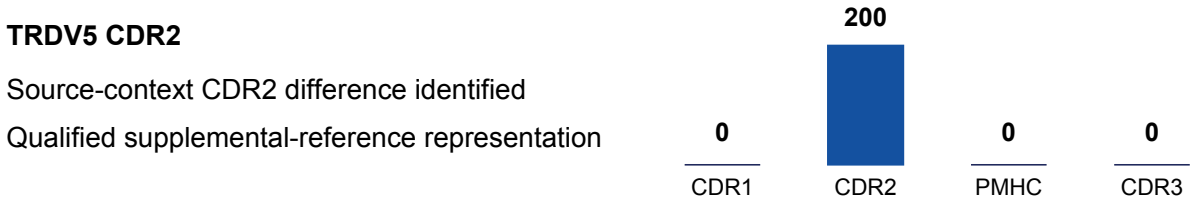
A Pinned mouse delta reference and admission



Separate J catalogue: 3 TRDJ alleles. Excluded primary records: 2.

TRAV21/DV12*03: ORF; TRAV6-7/DV9*08: missing CDR1.

B Source-context difference, not native scoring error



47 common alleles / 49 records; 200 directed totals differ only through CDR2.

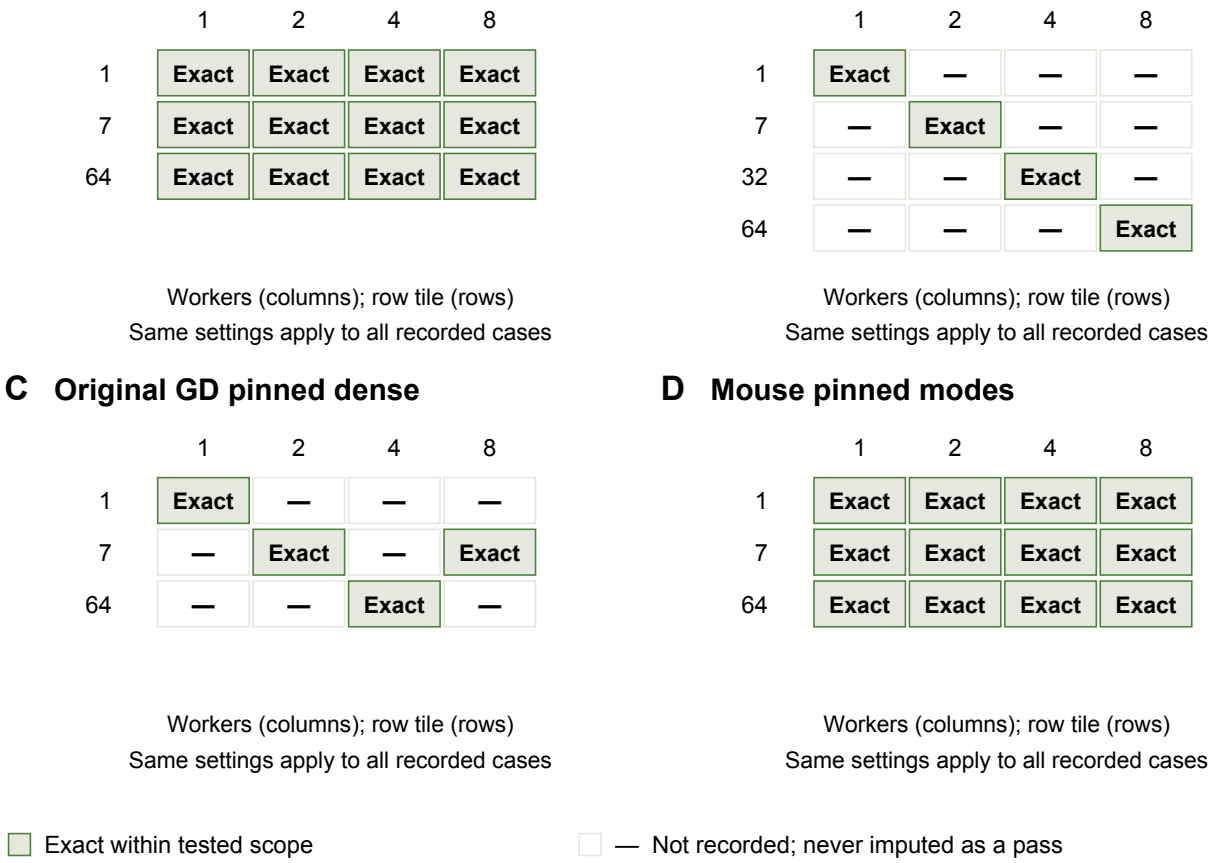
Feature-array intervals: 28–39 / 57–66 / 82–88, not generic IMGT numbering.

Source qualifiers F / (F) / [F] are retained; no independent function claim.

Archived IMGT source snapshot: 2026-10-29.

Mouse delta reference provenance. A shows 53 primary TRDV records, 51 admitted alleles (11 dedicated, 40 shared) and two exclusions; three TRDJ alleles are separate. B summarizes the TRDV5 CDR2 source-context difference. All 200 differing directed pair totals among 49 records (47 common alleles) are explained by CDR2; other regional discrepancies are zero. These are source-context differences, not native errors. Feature-array intervals are not generic IMGT numbering. Source qualifiers and namespace exclusions retain their stated limits (Methods).

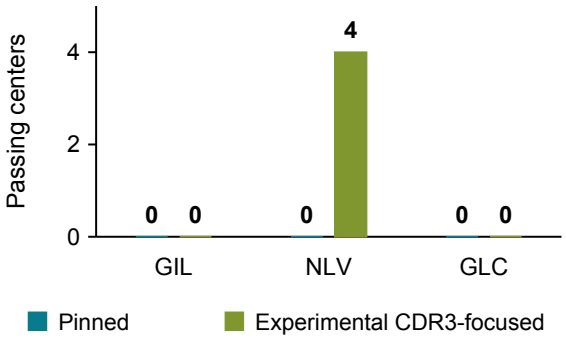
Supplementary Figure S2



Recorded determinism settings. A–D distinguish Core dense, Core radius-edge, original gamma/delta pinned dense and mouse pinned modes. Exact cells denote scientific equality within tested scope; dashes denote unrecorded settings, not imputed passes. Worker counts are columns and row tiles are rows. Output ordering, inclusive thresholds, explicit sparse zeros and conservation checks are described in Methods. Per-run metadata, other platforms, hard RSS limits and power-loss recovery remain outside this qualification.

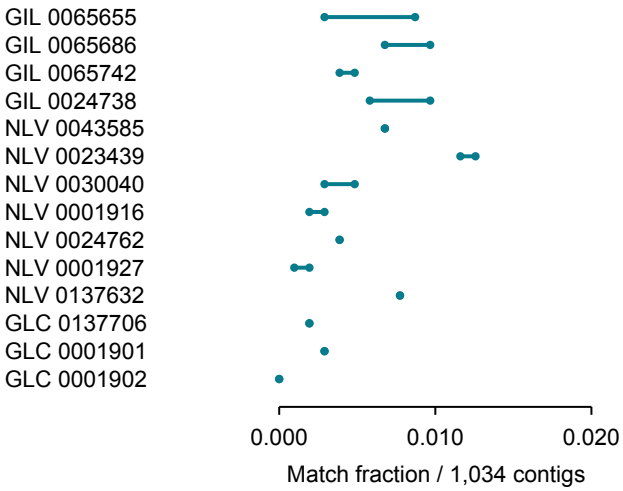
Supplementary Figure S3

A Reversed-label centers

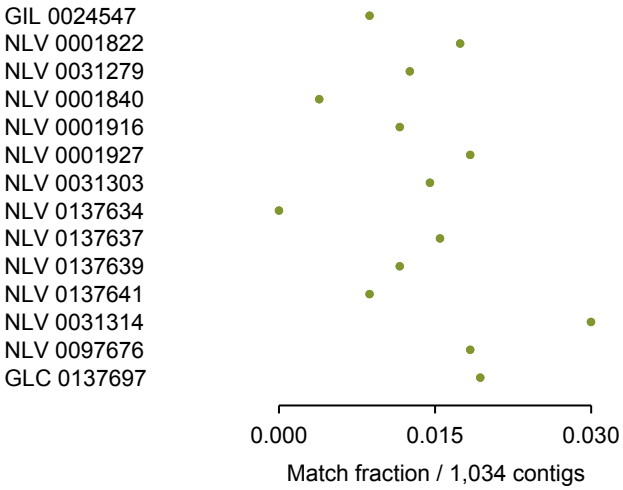


Four NLV experimental null centers persist.

B Pinned background bounds



C Experimental background bounds



D Computational controls

- **Synthetic positive: 16/16 accepted**
- **Distant negative: 0/16 accepted**

Leakage / seal tamper / wrong gene / contaminated zero-radius refusals; anchored motif control.

Controls do not establish biology.

Annotation bounds are not confidence intervals or assigned genotypes.

Controls and annotation sensitivity. GIL: GILGFVFTL, influenza A matrix protein epitope; NLV: NLVPMVATV, cytomegalovirus pp65 epitope; GLC: GLCTLVAML, Epstein-Barr virus BMLF1 epitope. A retains four experimental CDR3-focused NLV reversed-label centers; pinned controls retain none. B and C show 14 pinned and 14 experimental centers, respectively, against one unlabeled donor. Bounds divide certain/possible matches by 1,034 contigs; 2,945 reference-possibility rows are metadata, not the denominator. Bounds are not confidence intervals or genotype assignments; gene-constrained interpretation is unsupported. D shows synthetic accepted/evaluated counts and rule controls. These are computational controls, not biological or population-error validation. The alternative remains experimental; no untouched selection holdout remains.